AEC DRAWING INTELLIGENCE PLATFORM

DELIVERABLE 01 SPECIFICATION

System Architecture & Technology Stack Specification

DOC ID: DOC-AECDI-ARCH-01 **REVISION:** Rev 1.0 **DATE:** October 2026 **AUTHOR:** Sahil Navadiya
PROJECT: AEC Drawing Intelligence **ROLE:** AI Process Optimization Lead **TARGET:** Uppteam / ISP Global **SCOPE:** Architecture & Tech Stack Only

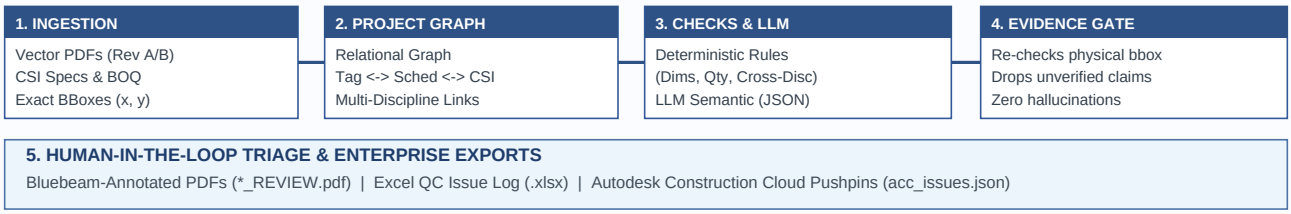
1.0. SYSTEM ARCHITECTURE & EXECUTION PIPELINE

? Core Principle: Extract & Structure -> Deterministic Rules Engine First -> Bounded LLM Semantic Reasoning -> Anti-Hallucination Evidence Gate -> Human Engineer Verification.

? Zero Hallucinations: No black-box PDF prompt chaining. 90%+ of checks execute via deterministic geometry, arithmetic, and cross-discipline graph rules.

? Physical Provenance: Every data point retains document name, revision, page number, and sub-millimeter bounding box coordinates (x0, y0, x1, y1).

END-TO-END DATA FLOW & EXECUTION PIPELINE



2.0. ARCHITECTURAL DECOMPOSITION (6 LAYERS)

2.1 Layer 1: Ingestion & Spatial Normalization

? Vector Drawing Parser: Extracts vector text spans, fonts, directional vectors, and exact bounding boxes using PyMuPDF (50+ sheets/sec, no OCR latency).

? Title Block & Index: Isolates sheet IDs (A-101, M-601), scales, and revision schedules. Reconciles sheet index vs actual sheets.

? Ruled Table Matrix: Extracts schedules (doors, windows, equipment, panels) by vector line intersections, mapping headers to cell grids.

? CSI Spec Parser: Splits project manuals into CSI MasterFormat sections, parts, articles, and lettered paragraphs with physical bounding boxes.

? BOQ Parser & Units: Scans Excel workbooks with openpyxl, preserving cell addresses (Sheet1!E18). Converts imperial fractions using exact rational arithmetic.

2.2 Layer 2: Multi-Discipline Relational Project Graph

? Property Graph: In-memory relational graph linking tags, schedules, specs, and BOQ items with typed edges (appears_on, defined_in, governed_by, quantifies, fed_by, supported_by).

Entity	Node Key	Relational Connections	Physical Provenance
Sheet	SHEET:A-101	references A-601, listed_in G-001	File, Page Index, Title Block BBox
Door Tag	DOOR:105	appears_on A-101, defined_in A-601	Plan Coordinate (x, y), Tag Shape
Door Schedule	DOORTYPE:D1	is_type DOOR:105, governed_by SPEC:08 14 16	Table Cell BBox, Schedule Row
Equipment Tag	EQUIP:RTU-1	appears_on M-101, fed_by CKT:HA-1, in BOQ	Plan BBox, Capacity, Weight
Electrical Ckt	CKT:HA-1	feeds EQUIP:RTU-1, on SHEET:E-601	Panel Name, Breaker Amps, Voltage
CSI Spec	SPEC:08 14 16	governs DOORTYPE:D1, refs SPEC:08 71 00	Section, Article, Paragraph BBox
BOQ Item	BOQ:08-110	quantifies DOORTYPE:D1, prices EQUIP:RTU-1	Sheet Name, Excel Row & Cell Address

2.3 Layer 3: Deterministic Rule Engines

? Dimensions (DIM): DIM-001: Chained vs overall sum (tolerance 1/8"). DIM-002: Vector length vs scale. DIM-003: Cross-sheet grid alignment (A-101 vs S-101).

? Quantities (QTY): QTY-001/2: Schedule vs BOQ counts. QTY-003: Flags NIC (Not In Contract) items in BOQ. QTY-004: Abstains on uncounted TYP. tags -> Needs Review. QTY-005: Plan tag multipliers vs schedule.

? Cross-Discipline (XD): XD-001: Mechanical V/PH vs Electrical panels (120V, 208V, 240V, 480V). XD-002: Unpowered equipment. XD-003: Breaker sizing vs MCA. XD-004: Equipment weight vs structural capacity.

? Completeness (MISS): MISS-001/2: Undefined / orphan tags. TAG-001: Duplicate tags. MISS-003: Blank mandatory schedule cells. MISS-004: Undefined hardware sets. MISS-005: Missing CSI spec sections.

? Revisions (REV): REV-001: Flags unclouded changes (3-layer diff: tables, text with shift compensation, geometry). REV-002: Changed content with unissued revision in title block.

2.4 Layer 4: Bounded LLM Semantic Layer (Claude Opus 5.5 / Sonnet)

? Target Scope: Reserved exclusively for subtle natural language conflicts between specifications and schedule notes.

? JSON Schema Enforcement: Strict schema requires element, attribute, schedule_value, paragraph_id, and verbatim spec_quote.

? Guardrails: Model forbidden from outside assumptions; output tier capped at 'Probable'; offline Replay & Mock clients for CI/CD.

2.5 Layer 5: Anti-Hallucination Evidence Gate

? Independent Verification: Re-opens source PDF/Excel, clips cited bounding box (+1.5pt pad), and verifies normalized snippet.

? Zero Tolerance: If snippet is missing or unverified, finding is dropped to rejected log. Shipped findings carry only 100% verified anchors.

2.6 Layer 6: Human-in-the-Loop & Enterprise Outputs

? Annotated Vector PDFs: Native vector rectangle annotations colored by severity (Red/Orange/Blue) for Bluebeam Revu.

? Excel QC Issue Log: Formatted workbook with frozen headers, auto-filters, and dropdown validations (Open, Accepted, Rejected, Deferred).

? Autodesk Construction Cloud: Formats issues for ACC Issues API with 2D vector centroid pushpins (x, y).

? Interactive Review App: Streamlit UI with side-by-side dynamic bounding box crops and append-only audit trail (review_log.jsonl).

3.0. FULL TECHNOLOGY STACK SPECIFICATION

Architecture Layer	Prototype Technology	Enterprise Target	Technical Role & Selection Rationale
Runtime / Core	Python 3.11+	Python 3.11+ / FastAPI	High throughput, rich scientific computing, robust typing.
Vector PDF Parser	PyMuPDF (fitz)	PyMuPDF + Azure Doc Intel	50+ vector sheets/sec. Azure provides fallback OCR for scans.
Table Matrix	Vector Line Intersection	PyMuPDF Tables + Textract	Deterministic cell extraction eliminates OCR errors on CAD tables.
Computer Vision Diff	OpenCV (headless) + NumPy	OpenCV + GPU Acceleration	Pixel overlay diffs (Red/Blue/Grey) in < 200ms at 150 DPI.
Spreadsheet Engine	openpyxl	openpyxl / DuckDB	Preserves exact cell coordinates (Sheet!E18) with low memory.
Imperial Math	Custom Fraction Engine	Custom Fraction Engine	Exact rational arithmetic eliminates floating-point rounding errors.
LLM Semantic Engine	Claude Opus 5.5 (Anthropic)	Claude 3.5 Sonnet / Llama-3	Zero Data Retention (ZDR) cloud API; local LLM for strict NDAs.
Workflow Automation	CLI Pipeline & Python	n8n (Self-Hosted) / Temporal	Event triggers on folder uploads (ACC, Dropbox) automate runs.
Review Dashboard	Streamlit 1.35+	React 19 + APS / PDF.js	Instant zero-overhead UI; React provides enterprise scale.
Database & Graph	JSON / SQLite	PostgreSQL + pgvector	Relational graph storage with ACID guarantees and vector search.